

2.8a

Intelligence & achievement Theories of Intelligence

Intelligence has often been defined as what intelligence tests measure, which has tended to be school smarts.

Intelligence is the ability to :

- learn from experience
- solve problems
- use knowledge to adapt

What is general intelligence?

Charles Spearman believed we have one general intelligence (g).

He noted that people scoring high in one area tended to score higher in others.

Spearman worked with factor analysis: a statistical procedure identifying clusters of related variables; which is where this belief came from.

What is the CHC theory?

Raymond Cattell & John Horn came up with a new general theory:

Cattell-Horn-Carroll theory.

CHC theory based on 2 factors:

fluid intelligence G_f : ability to reason abstractly ex. w/ logic puzzles

Crystallised intelligence G_c : accumulated knowledge, ex. vocabulary

Our G_f & G_c often work together

Do we have multiple intelligences?

Howard Gardner identified 8 relatively independent intelligences

this incl. verbal & math skills measured by standardized tests.

this idea has influenced pseudo-scientific notions of 'learning styles'.

Brain damage may destroy one kind of intelligence, while leaving others intact.

People with Savant Syndrome have limited to no language ability, & score low on intelligence tests.

But they have amazing abilities, ex.:

- mentally computing complex calculations
- identifying date of week for some historical date.
- rendering incredible art works

Sternberg's 3 intelligences

Sternberg's triarchic theory proposes 3 reliably measured intelligences

- Analytical intelligence: predicts school grade well; assessed by intelligence tests
- Creative intelligence: ability to adapt to new situations and generate novel ideas
- Practical intelligence: for every day tasks that are poorly defined & may have multiple solutions

General Intelligence & Success

g predicts performance on complex tasks & jobs

g predicts higher income

extremely high cognitive ability scores predict exceptional accomplishments

Success is a combination of talent & grit.

expert performance in various skills takes ~10 yrs

What is emotional intelligence?

Social Intelligence is understanding social situations & managing self successfully.

Emotional Intelligence, part of SI, consists of 4 abilities:

- Perceiving Emotions (recognizing them in faces, music, etc.)
- Understanding emotions (predicting them & how they may change)
- Managing emotions (knowing how to express them & how to handle others')
- Using emotions (to facilitate creative or adaptive thinking)

Personal Connection:

While I have many issues with the idea of intelligence, If I had to use it, I would say that my strengths lie in academic intelligence, and, in areas I specialize in where it is applicable like politics, philosophy & science, creative intelligence. Not as much in emotional intelligence.

MCQ

1. c 2. c 3. d 4. b 5. a